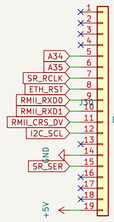
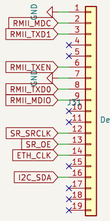
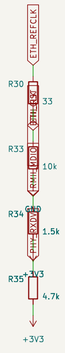


12 V in: reverse protection D25, TVS D26, LM2596 buck to 5 V, AMS1117 to 3.3 V. D28 lets the DevKitC USB power the logic for bench tests.

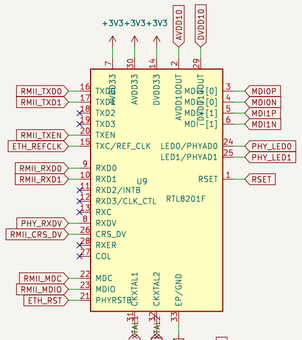


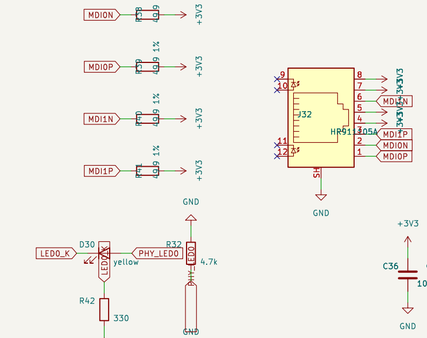


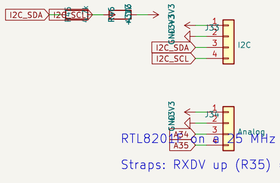


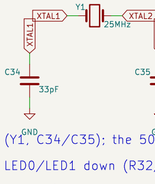
ESP32 DevKitC V4 (38 pins) in two 1x19 sockets. RMII pins are fixed by the ESP32: TXD0 19, TXD1 22, TXEN 21, RXD0 25, RXD1 26, CRS_DV 27, MDC 23, MDIO 18, 50 MHz reference clock in on GPIO0 (from the PHY).

Shift registers: SER 13, SRCLK 16, RCLK 32, OE 4 (pulled up: all relays off until the program starts). PHY reset 33. I2C 15/14. Analog 34/35.





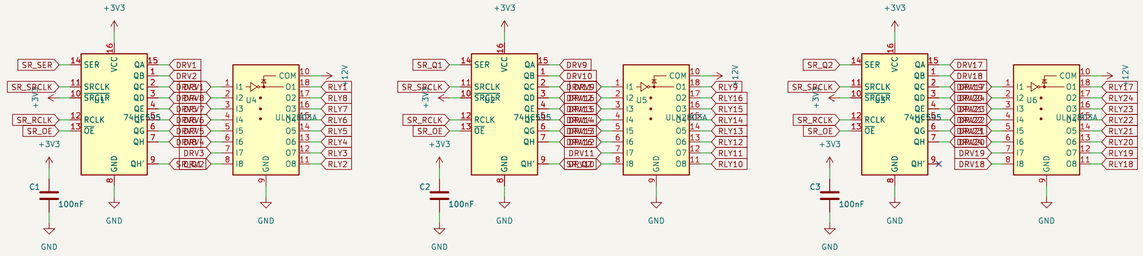




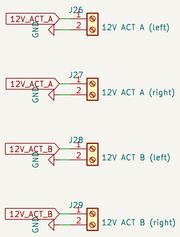
RTL8201F on a 25 MHz crystal (Y1, C34/C35); the 50 MHz RMII clock comes OUT of TXC -> R30 -> GPIO0 (ETH_CLOCK_GPIO0_IN). R33 keeps the PHY in reset until GP

Straps: RXDV up (R35) = RMII; LED0/LED1 down (R32/R36) = PHY address 0, LEDs active high: D30 = 10 Mbit link/activity, D31 = 100 Mbit (default LED mode). RSET

RJ45 HR911105A: PHY pair 0 (transmit) on pins 1–2, pair 1 (receive) on 3–6, centre taps (4, 5, 7, 8) to 3.3 V with C36/C37, its own LEDs unused.



Drivers: 74HC595 (3.3 V) -> ULN2803A (coils at 12 V, flyback diodes inside, COM to +12V). One ULN per sheet of 8 relays.



Relays 1–8 (sections 1–8, driver U4)

File: relays_1_8.kicad_sch

Relays 9–16 (sections 9–16, driver U5)

File: relays_9_16.kicad_sch

Relays 17–24 (sections 17–24, driver U6)

File: relays_17_24.kicad_sch



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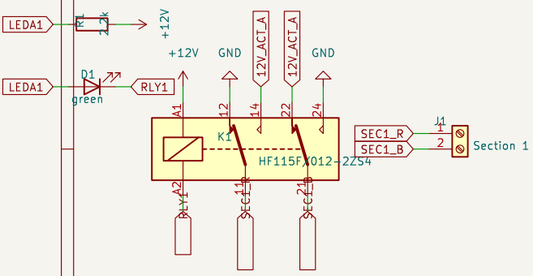
renatorgr

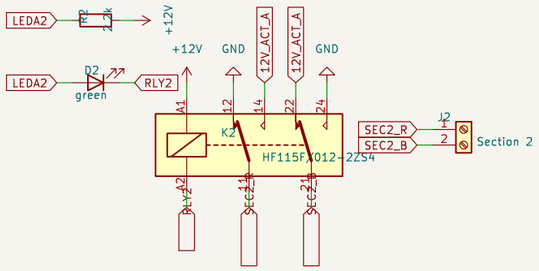
Sheet: /

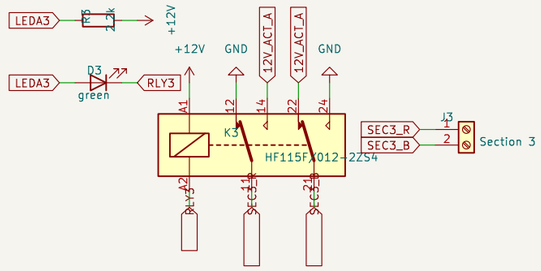
File: Relay24_ESP32.kicad_sch

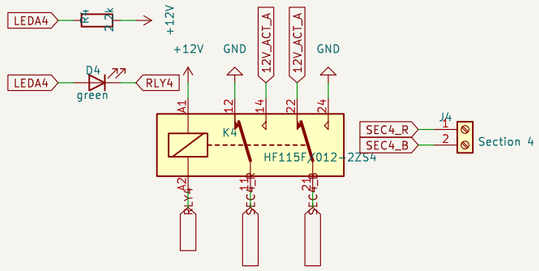
Title: Relay24_ESP32 – 24-section relay board for an ESP32 DevKitC (AgOp

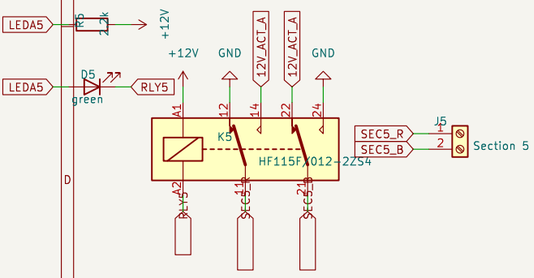
Size: A2	Date: 2026-10-28	**Rev: 1.0**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/4

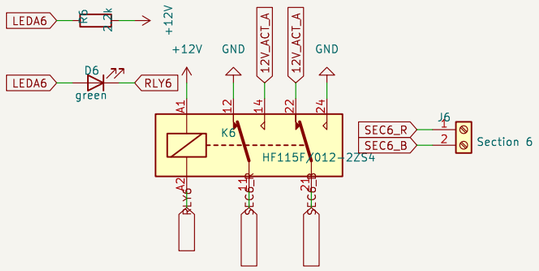


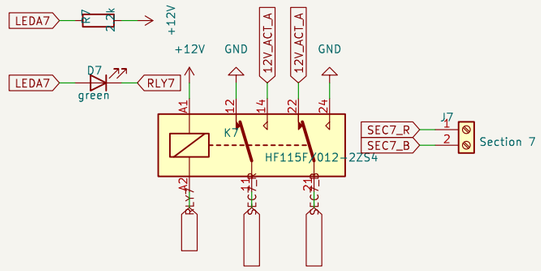


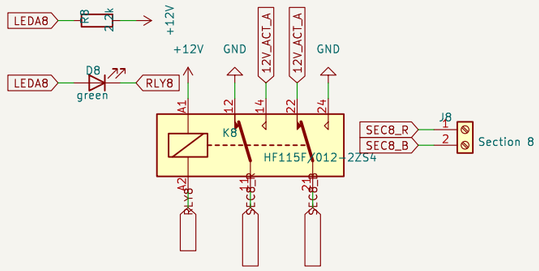












Each section: relay K (DPDT, coil A1 +12V / A2 driven low by the ULN), terminal J (1 = red wire, 2 = black wire). Relay off: red at GND, black at +12V ACT (pull in, closed). Relay on: red +12V ACT, black GND (push out, open).

LED D with R 2.2k across the coil shows the section open. 12V_ACT_A = sections 1–12 (J26/J27), 12V_ACT_B = sections 13–24 (J28/J29).

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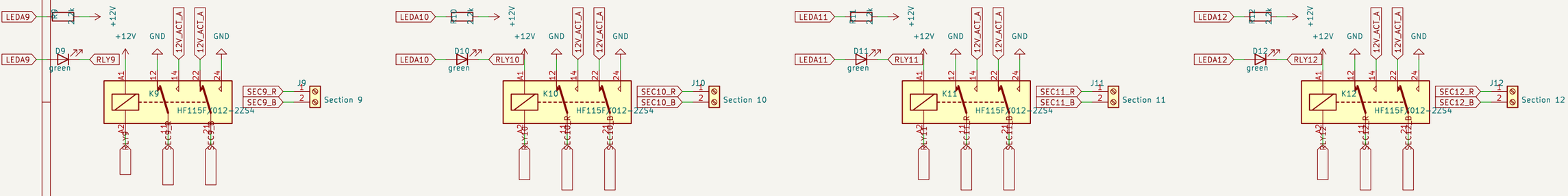
Sheet: /Relays 1–8 (sections 1–8, driver U4)/

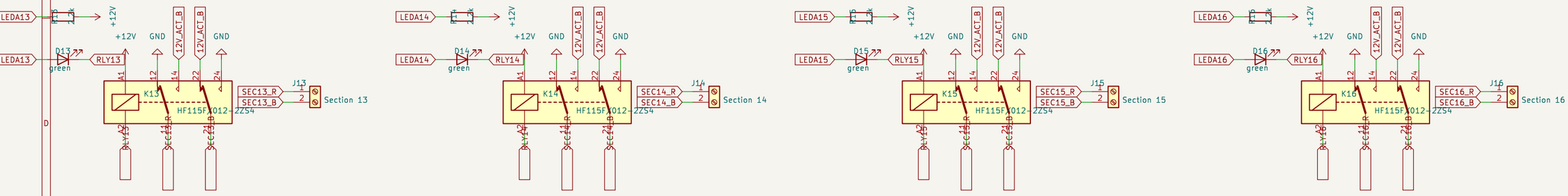
File: relays_1_8.kicad_sch

Title: Relays 1–8 (sections 1–8, driver U4)

Size: A3 | Date: 2026-10-28 | **Rev: 1.0**

KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 | Id: 2/4





Each section: relay K (DPDT, coil A1 +12V / A2 driven low by the ULN), terminal J (1 = red wire, 2 = black wire). Relay off: red at GND, black at +12V ACT (pull in, closed). Relay on: red +12V ACT, black GND (push out, open).

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Written by Claude (Anthropic) for renatorgr, GPL v3 or later

renatorgr

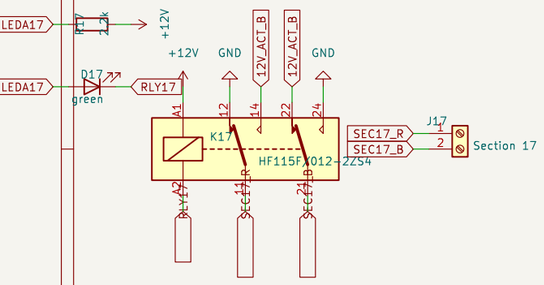
Sheet: /Relays 9–16 (sections 9–16, driver U5)/

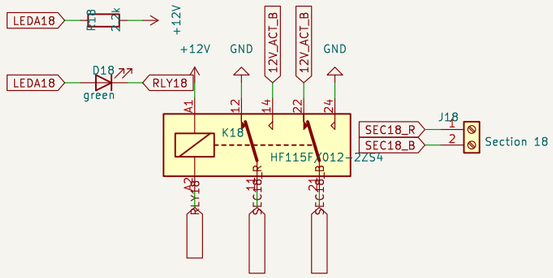
File: relays_9_16.kicad_sch

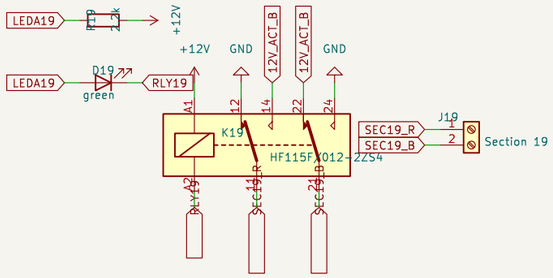
Title: Relays 9–16 (sections 9–16, driver U5)

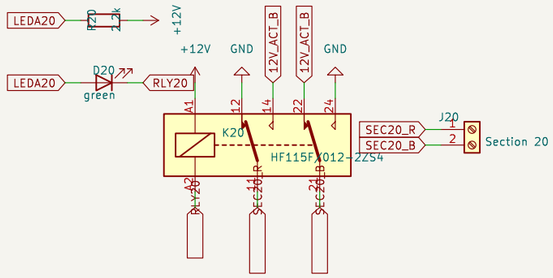
Size: A3 | Date: 2026-10-28 | **Rev: 1.0**

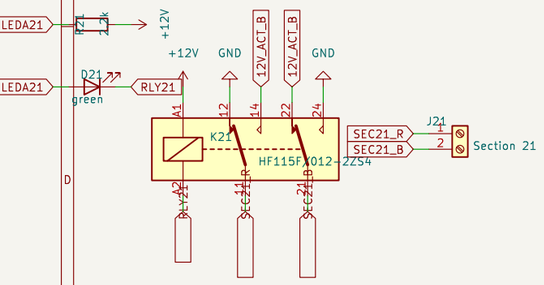
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 | Id: 3/4

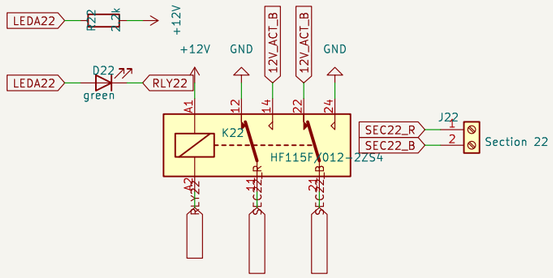


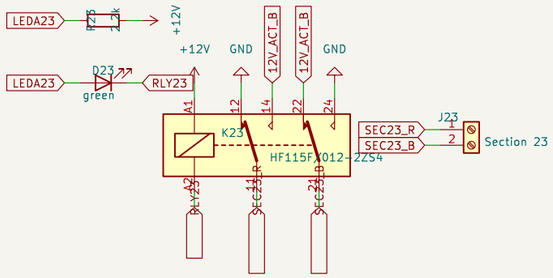


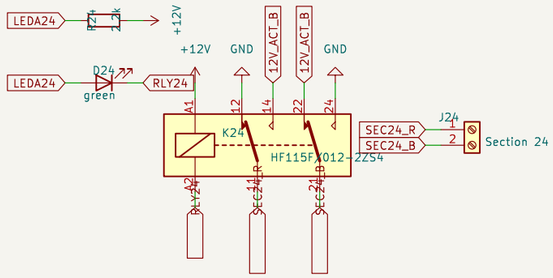












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LED D with R 2.2k across the coil shows the section open. 12V_ACT_A = sections 1–12 (J26/J27), 12V_ACT_B = sections 13–24 (J28/J29).

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Sheet: /Relays 17–24 (sections 17–24, driver U6)/

File: relays_17_24.kicad_sch

Title: Relays 17–24 (sections 17–24, driver U6)

Size: A3 | Date: 2026-10-28 | **Rev: 1.0**

KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 | Id: 4/4